Concepts Make Confidence

MEDICOS MINDS

LEARN PRACTICE EXCEL

BIOCHEMISTRY

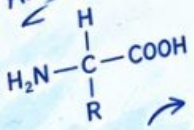
THE COMPLETE STUDY GUIDE

FOR MBBS | BHMS | B.Sc. NURSING | BDS | BPT

Small Concepts Big Impact



Amino Acids





Carbohydrates



Energy (ATP)

Lipids



Proteins

Proteins



KEY AREAS

- ✓ Carbohydrates
- ✓ Proteins
- ✓ Lipids
- ✓ Enzymes
- ✓ Vitamins
- ✓ Nucleic Acids
- ✓ Metabolism
- ✓ Molecular Biology
- ✓ Clinical Biochemistry

WHY THIS BOOK?

- High-Yield Points
- Concept Clarity
- Diagrams & Flowcharts
- Previous Year Exam Focus
- Easy to Revise

Biochemistry Builds the Foundation of Life

Biochemistry = Life

CARBOHYDRATES & LIPIDS | PROTEINS & AMINO ACIDS | ENZYMES & CATALYSIS | VITAMINS & MINERALS | NUCLEIC ACIDS & GENETICS | METABOLISM & BIOENERGETICS

Better Biochemistry Better Care

@restore_healthphysio

Study Smart Grow

BIOCHEMISTRY

MBBS
BHMS
B.Sc. NURSING
BDS
BPT

MEDICOS MINDS

HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Biochemistry

Author: Dr. MD Kamran Alam | Instagram: @restore_healthphysio

OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 59

Authorized License: Personal Single-User Academic License

STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

2.

1. pH and Buffers :-

* Definitions :-

1. Acid :- Substance that donates H^+ ions in solution.
   Ex. HCl, H_2SO_4
2. Base :- Substance that accepts H^+ ions or releases OH^- ions in solution.
   Ex. $NaOH$, NH_4OH

* Strong and Weak Acids / Bases :-

- Strong acids :- Completely dissociate in water to release H^+ ions.
  Ex. HCl, H_2SO_4, HNO_3
- Weak acids :- Partially dissociate in water.
  Ex. CH_3COOH, H_2CO_3
- Strong bases :- Completely dissociate to release OH^- ions.
  Ex. $NaOH$, KOH
- Weak bases :- Partially dissociate in water.
  Ex. NH_4OH, NH_3

* Henderson-Hasselbalch Equation :-

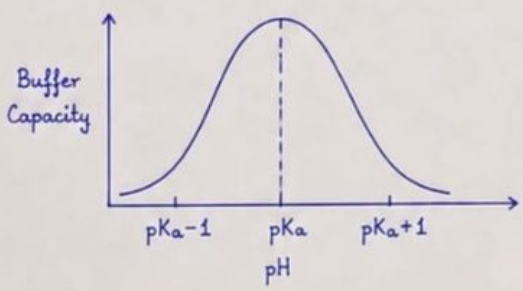
It is used to calculate pH of buffer solution.

$$pH = pKa + \log \frac{[Salt]}{[Acid]}$$

Where,
pKa = $-\log$ Ka (Acid dissociation constant)
[Salt] = Concentration of salt (conjugate base)
[Acid] = Concentration of weak acid

* pKa and Buffer Capacity :-

- pKa is the pH at which acid and its salt are present in equal concentration.
- Buffer capacity is maximum when pH = pKa.
- Buffer capacity decreases when pH deviates 1 unit above or below pKa.



* Buffer Capacity :-

- It is the ability of a buffer to resist change in pH when small amounts of acid or base are added.
- Depends on concentration of buffer components.
- Maximum buffer capacity at pH = pKa.

* Normal pH of Blood and its Regulation :-

- Normal pH of arterial blood : 7.35 – 7.45 (slightly alkaline)
- pH < 7.35 ⟶ Acidosis
- pH > 7.45 ⟶ Alkalosis
- pH is maintained by:
  1. Buffer systems (Immediate)
  2. Respiratory system (Minutes)
  3. Renal system (Hours to days)

* Respiratory Regulation (Lungs) :-

- Lungs regulate pH by controlling amount of CO_2 in blood.
- $CO_2 + H_2O \underset{\text{(Carbonic anhydrase)}}{\rightleftharpoons} H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$
- ↑ Ventilation ⟶ ↓ CO_2 ⟶ ↑ pH (Alkalosis)
- ↓ Ventilation ⟶ ↑ CO_2 ⟶ ↓ pH (Acidosis)

* Renal Regulation (Kidneys) :-

- Kidneys regulate pH by:
  1. Excreting H^+ ions in urine.
  2. Reabsorbing HCO_3^- (bicarbonate).
  3. Generating new HCO_3^- in renal tubules.
- Important in long term regulation (hours to days).

* Buffer Systems in Body Fluids :-

Buffer System	Components	Function
1. Bicarbonate Buffer System	H_2CO_3 / HCO_3^-	Main extracellular buffer, regulated by lungs & kidneys.
2. Phosphate Buffer System	$H_2PO_4^-$ / HPO_4^{2-}	Important intracellular buffer.
3. Protein Buffer System	Protein- / Protein-H	Present in all cells and body fluids.
4. Hemoglobin Buffer System	Hb / HHb	Important in RBCs, helps in CO_2 transport and buffering.

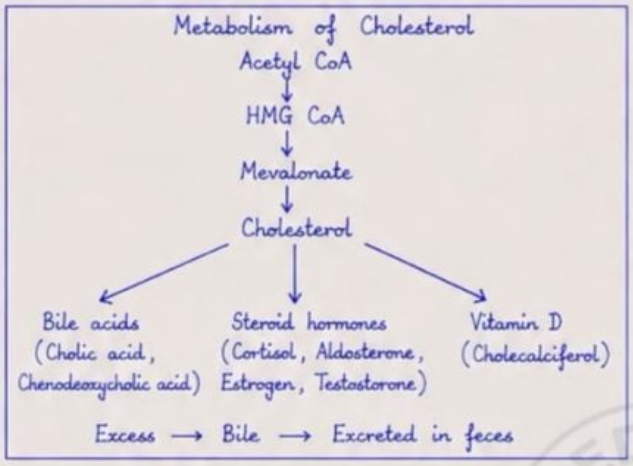
Note :- All buffer systems act immediately to resist change in pH.

10.

4. Lipids :-

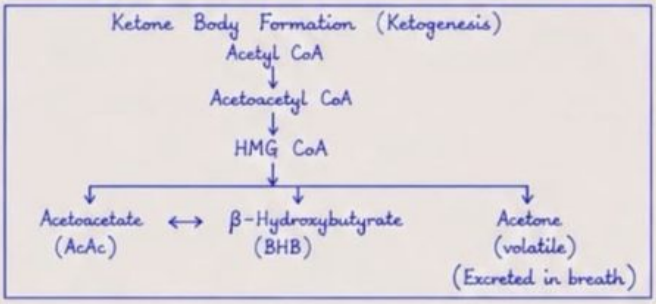
1. Metabolism of Cholesterol :-

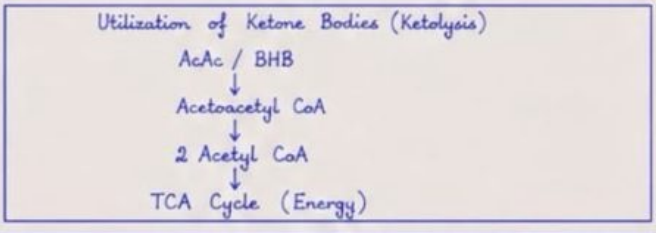
- Cholesterol is synthesized mainly in liver, intestine and transported to tissues.
- Acetyl CoA is the precursor.
- Major end products : Bile acids, Steroid hormones, Vitamin D.
- Excess cholesterol is excreted in bile as cholesterol or bile acids.



2. Ketone Body Metabolism :-

- Ketone bodies are produced in liver from acetyl CoA during fatty acid oxidation (especially in starvation, prolonged fasting, uncontrolled diabetes).
- Ketone bodies : Acetoacetate, β-Hydroxybutyrate, Acetone.
- They are used by extrahepatic tissues (muscle, heart, brain during starvation) as an alternative energy source.

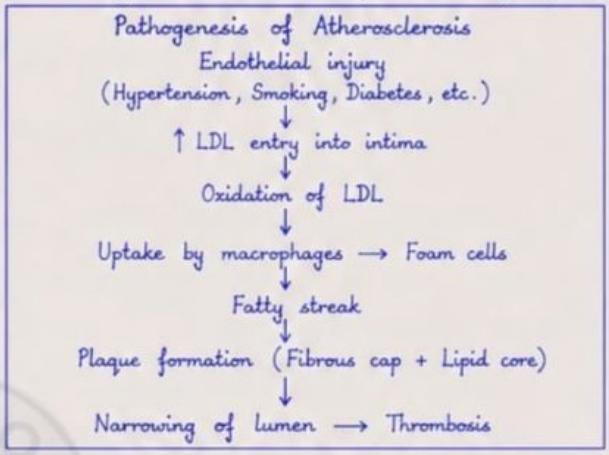


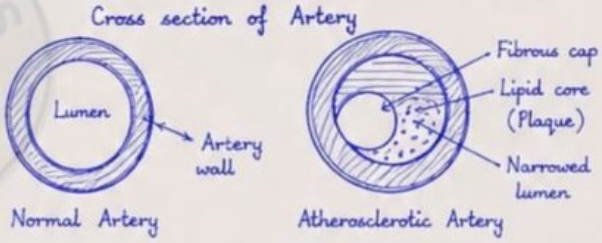


Key Points	
Site of formation	Liver (mitochondria)
Site of utilization	Muscle, heart, kidney, brain (starvation) (not in liver)
Transport form	As ketone bodies in blood
Physiological role	Provides energy during carbohydrate deficiency

3. Atherosclerosis :-

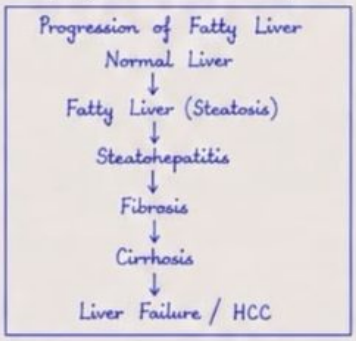
- Atherosclerosis is a condition in which arteries become narrowed and hardened due to deposition of lipids, cholesterol and fibrous tissue in the intima.
- It reduces blood flow and may lead to heart attack, stroke or peripheral vascular disease.
- Risk factors : High LDL, Low HDL, Hypertension, Smoking, Diabetes, Obesity, Sedentary life style.

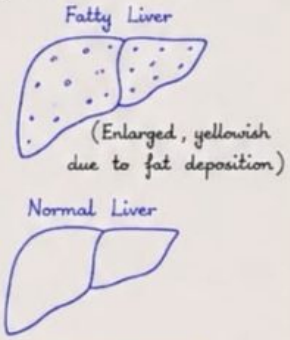




4. Fatty Liver (Hepatic Steatosis) :-

- Fatty liver is the accumulation of excessive fat (triglycerides) in liver cells.
- Causes : Alcohol abuse, Obesity, Diabetes mellitus, High fat diet, Malnutrition, Some drugs.
- It may be reversible (simple steatosis) or may progress to steatohepatitis, fibrosis, cirrhosis.





Key Points	
Major lipid stored	Triglycerides
Site	Liver cytoplasm
Reversible	Yes (in early stage)
Complications	Cirrhosis, Liver failure, HCC

THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

SUPPORT & COMMUNITY:

- Instagram: @restore_healthphysio (Daily Clinical Case Studies & Quizzes)
- WhatsApp Helpline: +91 83407 49923
- Email Support: restorehealthphysio@gmail.com

MEDICOS MINDS • All Rights Reserved • Confidential & Educational Material